Midterm Project Group 8

Group members: Ryan Merlino, Nikolas Platon, April Walter

“How is the number of diagnoses related to hospital readmission, and does that relationship differ by admission type (Emergency

age str	gender str	admission_type str	time_in_hospital int64	num_lab_procedures int64	num_procedures int64	num_medications int64	number_diagnoses int64	diab str
70-79	Male	Elective	4	27	4	23	5	Yes
80+	Female	Elective	11	70	5	35	9	No
60-69	Male	Emergency	5	69	0	18	9	Yes
70-79	Female	Emergency	5	73	0	19	9	Yes
60-69	Male	Urgent	5	37	0	16	5	Yes

5 rows, 10 columns

No selection

```
(3000, 10)
age                    str
gender                 str
admission_type         str
time_in_hospital     int64
num_lab_procedures   int64
num_procedures       int64
num_medications      int64
number_diagnoses     int64
diabetes_med           str
readmitted             str
dtype: object
Missing values: 0
```

admission_type str	number_diagnoses int64	readmitted str	readmitted_flag int64	diagr str
Elective	5	Not readmitted	0	5-6
Elective	9	Within 30 days	1	9+
Emergency	9	Within 30 days	1	9+
Emergency	9	Not readmitted	0	9+
Urgent	5	Not readmitted	0	5-6

5 rows, 5 columns
No selection

diagnosis_group str	readmitted_flag float64
1-4	24.4
5-6	34.6
7-8	47.3
9+	46.3

4 rows, 1 column
No selection

```
All patients:       3000
Emergency patients: 1716
```

admission_type str	readmitted_flag float64
Elective	34.1
Emergency	41.8
Urgent	41.4

3 rows, 1 column
No selection

Showing the first **10** of **12** rows. Increase the table's `page_size` to embed more in the static export.

admission_type str	diagnosis_group str	readmission_rate float64
Elective	1-4	21.0
Elective	5-6	27.0
Elective	7-8	36.6
Elective	9+	42.2
Emergency	1-4	24.4
Emergency	5-6	34.6
Emergency	7-8	47.3
Emergency	9+	46.3
Urgent	1-4	29.7
Urgent	5-6	30.7

12 rows, 3 columns

No selection

Showing the first **10** of **12** rows. Increase the table's `page_size` to embed more in the static export.

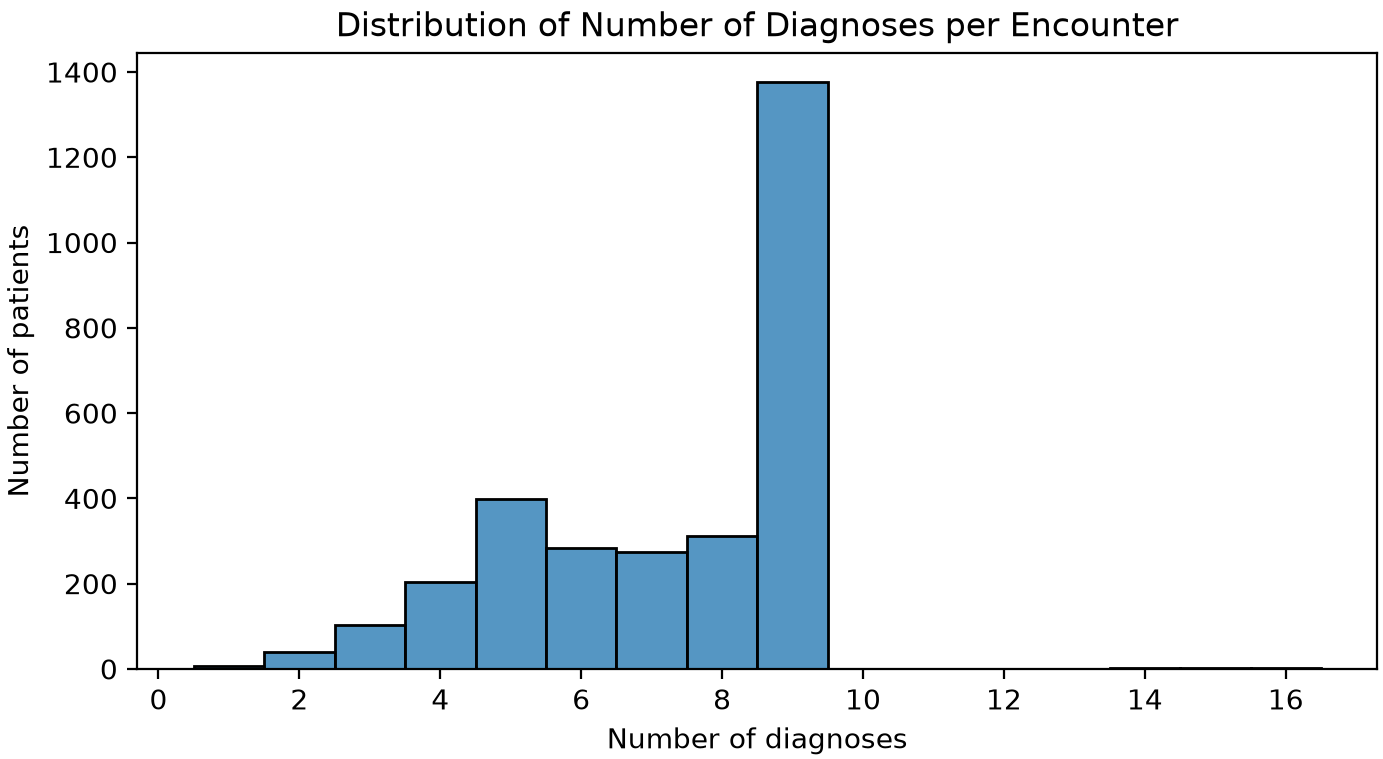
Index0 int64	admission_type str	diagnosis_group str	readmission_rate float64
11	Urgent	9+	48.4
10	Urgent	7-8	48.0
6	Emergency	7-8	47.3
7	Emergency	9+	46.3
3	Elective	9+	42.2
2	Elective	7-8	36.6
5	Emergency	5-6	34.6
9	Urgent	5-6	30.7
8	Urgent	1-4	29.7
1	Elective	5-6	27.0

12 rows, 3 columns

No selection

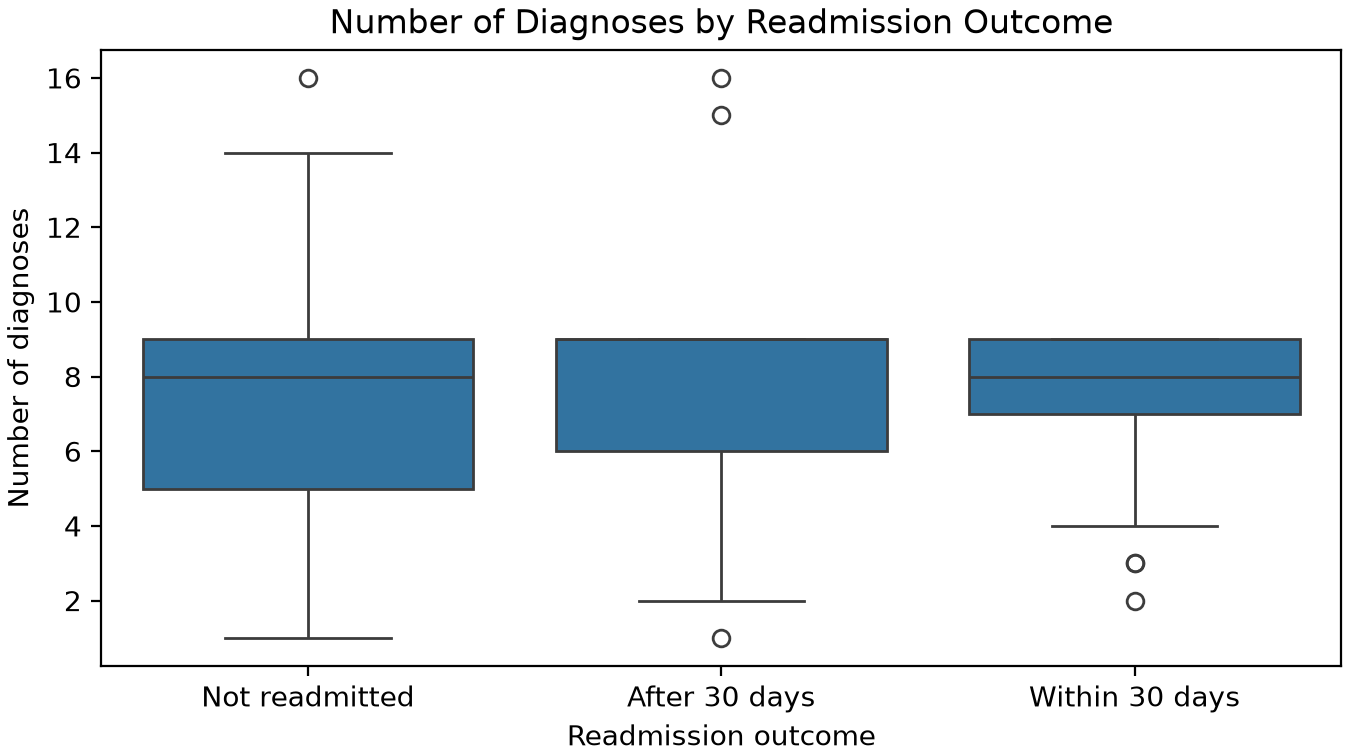
Histogram- Distribution of Number of Diagnoses per Encounter

We use a histogram because it shows how many patients have each number of diagnoses. This helps us see which numbers are c
rare before we compare readmission. It is also why we put the diagnoses into groups and leave out the rare values in the scatter



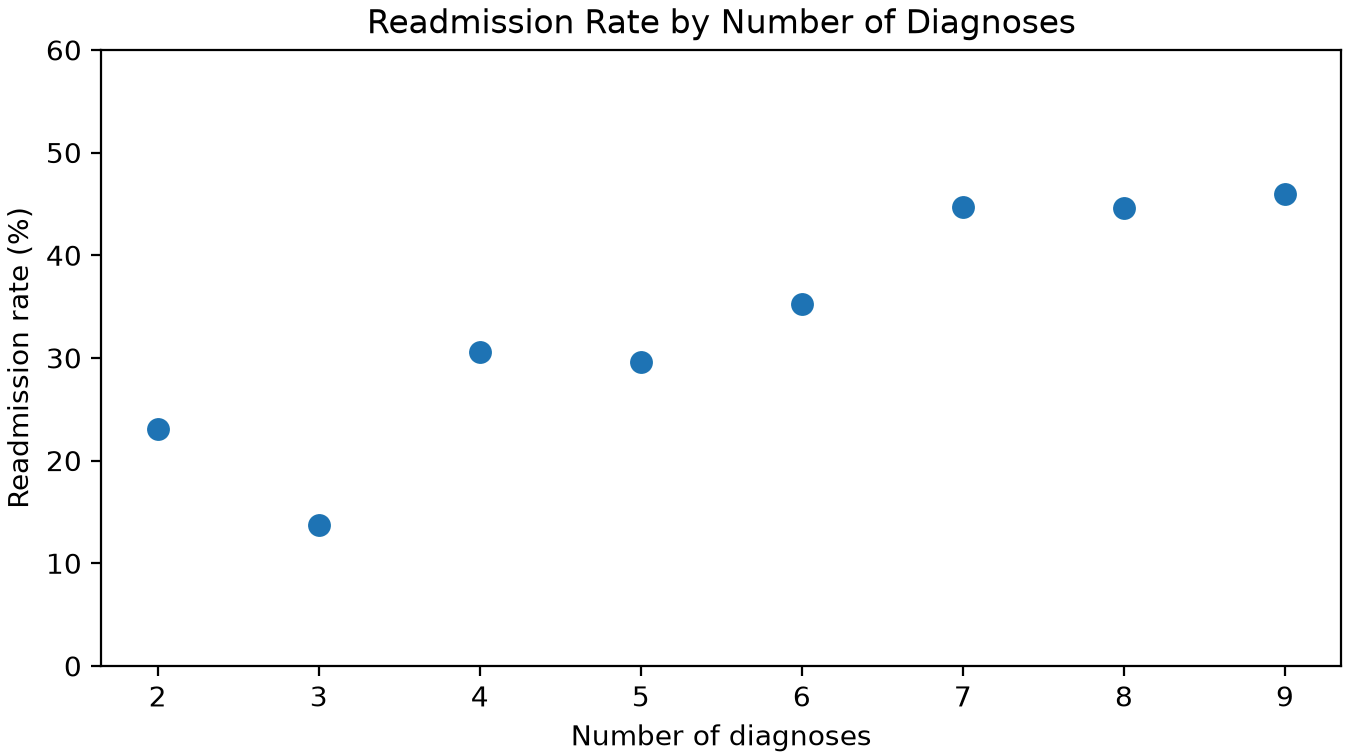
Boxplot- Number of Diagnoses by Readmission Outc

We use a boxplot because it lets us compare the number of diagnoses across the three readmission outcomes. We are looking to
readmitted tend to have more diagnoses than patients who were not readmitted.



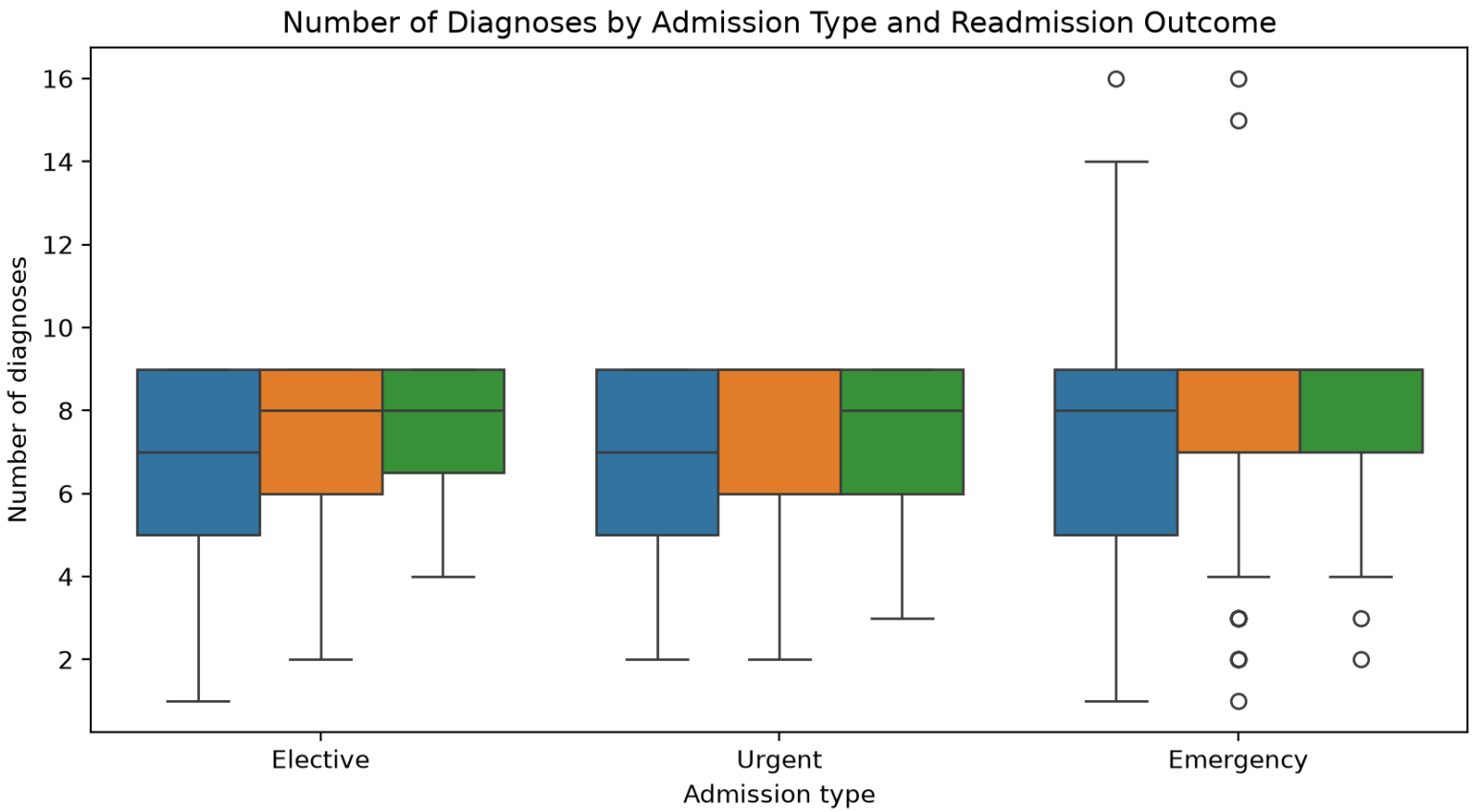
Scatterplot- Readmission Rate by Number of Diagno

We use a scatter plot because it shows how two numbers move together, here the number of diagnoses and the readmission rate the readmission rate goes up as the number of diagnoses goes up. We only show 2 to 9 diagnoses because the other values have diagnosis has 6 patients and 14 to 16 have only 1 or 2 each), so their rates would not be reliable.



Group Comparison- Number of Diagnoses by Admission Type and Readmission Outcome

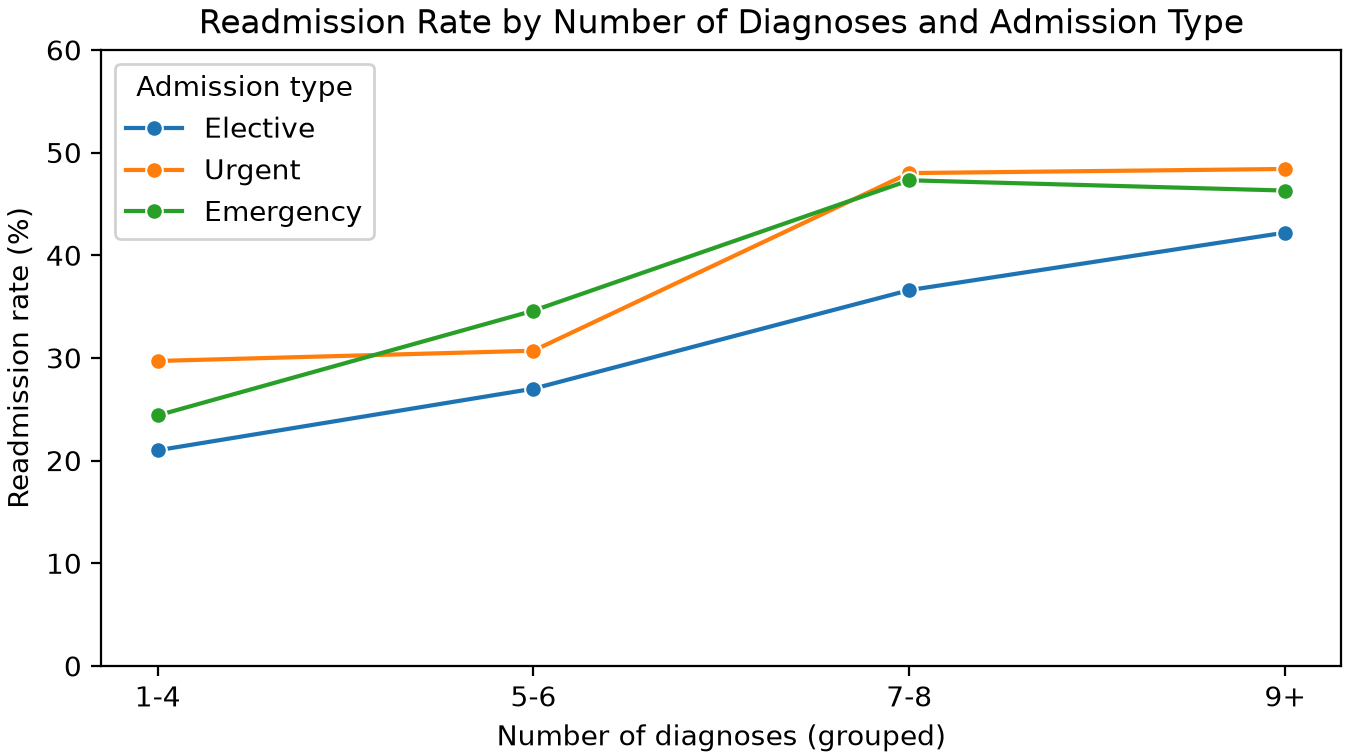
We use a grouped boxplot so we can compare readmission outcomes within each admission type side by side. We are looking to diagnoses between readmitted and not readmitted patients is the same for Emergency, Urgent and Elective.



Line Chart- Readmission Rate by Number of Diagnoses and Admission Type

We use a line chart because the diagnosis groups are in order from low to high, so a line shows the trend. Each admission type g

lets us see if readmission rises the same way for all three types or if one is different. This chart answers the second part of our re



Findings

• 40.0% of all encounters were followed by a readmission at any time.

• The readmission rate rises from 24.6% for patients with 1–4 diagnoses to 31.9% (5–6), 44.6% (7–8) and 46.0% (9+). It levels of

• Elective admissions had a lower readmission rate (34.1%) than Urgent (41.4%) and Emergency (41.8%), and Elective is the lowe group.

• The Emergency-only filter (Step 6) shows the same rising pattern within a single admission type.

Limitations

• This shows a relationship, not a cause. More diagnoses may mean sicker patients, and that could be what drives readmission.

• Most encounters have exactly 9 diagnoses, because the original hospital systems stored at most 9 for most of these years. Patie conditions may be undercounted.

• The data is a 3,000-row sample from 1999–2008 covering only patients with diabetes, so results may not apply to other patien

• Some combinations are small. Urgent with 1–4 diagnoses has only 74 patients, so its rate is less certain.